

Problem 4

A body moves along a straight path such that in successive equal time intervals of duration Δt , it maintains constant speeds $v_1, v_2, v_3, \dots, v_n$. What is the average speed of the body?

Solution

To calculate the average speed when time intervals are equal, we use the formula:

$$v_{\text{avg}} = \frac{v_1 + v_2 + v_3 + \dots + v_n}{n},$$

where n is the number of equal time intervals.

Steps

1. For two time intervals ($n = 2$):

$$v_{\text{avg}} = \frac{v_1 + v_2}{2}.$$

2. For three time intervals ($n = 3$):

$$v_{\text{avg}} = \frac{v_1 + v_2 + v_3}{3}.$$

3. In the general case for n time intervals:

$$v_{\text{avg}} = \frac{\sum_{i=1}^n v_i}{n}.$$

Conclusion

The average speed of the body is the arithmetic mean of the speeds over the equal time intervals.